

Break in -ve wire of cable $\Rightarrow$ voltmeter reading = 24 V when connected to X or Y

21 A In dark condition:

$$V = \frac{5 \times 10^3}{5 \times 10^3 + 5 \times 10^6} \times 6 = 0.0060 \text{ V}$$

In bright condition:

$$V = \frac{5 \times 10^3}{5 \times 10^3 + 1 \times 10^3} \times 6 = 5.0 \text{ V}$$

22 C

D